

One-Channel Mixed-Signal

Impedance Front End

Project	Senior Design: Low-Power Agricultural Impedance Sensing
Architecture	Fixed-gain TIA, 8-bit SAR ADC, and minimal SPI backend
Team	Krishna, Taara, and Vidhu
Revision	3.0 – Three-Person Minimal-Hardware Baseline
Date	September 1, 2026
Status	Simplified implementation specification; circuit values remain subject to simulation

This revision reduces the original four-channel autonomous measurement system to one externally excited channel. The ASIC retains a meaningful mixed-signal contribution while moving waveform generation, synchronous demodulation, averaging, and calibration to the MCU.

Document Purpose

This document defines a reduced-scope ASIC that can be completed and verified by Krishna, Taara, and Vidhu within the senior-design schedule. The chip senses one electrode current with a transimpedance amplifier (TIA), digitizes the TIA output with an 8-bit successive-approximation (SAR) ADC, stores the result, and returns it to an external microcontroller through SPI. The MCU and board generate the excitation waveform and perform only the simple paired-sample arithmetic needed for demonstration.

The primary source remains the submitted proposal, *Mixed-Signal Impedance Sensing ASIC for Battery-Free Agricultural IoT Nodes*. The proposal establishes the agricultural sensing use case, the impedance-measurement motivation, duty-cycled operation, and MCU connectivity. This revision intentionally removes several proposed blocks to match the available team and schedule. It should therefore be treated as the current implementation baseline.

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1 Executive Summary

The revised ASIC is a small, one-channel mixed-signal data-acquisition front end. An external MCU drives a low-frequency bipolar excitation across a soil or plant sensor. The return current enters the ASIC's TIA, which converts it into a voltage. On command, an on-chip 8-bit SAR ADC samples that voltage and stores the digital code. The MCU reads the code through SPI and forwards structured measurements to a board-side test script when needed.

To recover a response correlated with the excitation, the MCU takes samples during both excitation polarities and a simple board-side script subtracts or averages them. This supporting code is only a test utility; it is not a separate project workstream. The chip remains centered on three tapeout-relevant blocks: the TIA, the SAR analog core, and the small digital backend.

Table 1: Revised architecture at a glance.

Decision	Baseline	Reason
Sensor channels	One direct current-sense input	Removes the analog MUX, channel registers, and scanning logic.
Excitation	Generated externally by the MCU/board	Removes the excitation DAC and on-chip frequency dividers.
Analog front end	One fixed-gain TIA	Retains the central low-current circuit challenge with a small interface.
Conversion	8-bit SAR ADC	Provides a genuine mixed-signal backend with manageable resolution.
Digital backend	SAR controller, result/status registers, and SPI	Enough digital content to control and read the ADC without a large system FSM.
Signal processing	Simple external subtraction, averaging, and calibration	Keeps demodulation and DSP off chip so the three-person team can finish the hardware.
References	External V_{REFH} , V_{REFL} , and V_{CM}	Removes the on-chip bandgap and simplifies first-silicon debugging.
Clocking	External digital clock; no PLL	Predictable, low-risk clock source for the SAR controller.

2 Scope and Requirements

2.1 Blocks included in the ASIC

- electrode return/current-sense input and appropriate pad protection;
- fixed-gain TIA with feedback resistor R_F and capacitor C_F ;
- 8-bit SAR ADC analog core: sampling network/CDAC, comparator, and reference switches;
- 8-bit SAR controller;
- one result register plus small control and status registers;
- SPI slave interface;
- reset, enable, data-ready, and basic bias-shutdown control;

- analog test access for the TIA output and ADC input path, subject to pad budget.

2.2 Functions assigned to the MCU and board

- generate the sensor excitation waveform;
- define excitation amplitude and frequency;
- provide the ASIC clock and external reference/common-mode voltages;
- align conversion commands with excitation polarity;
- read ADC codes over SPI;
- optionally package raw samples with excitation state and timing metadata;
- run simple subtraction, averaging, and calibration during board bring-up.

2.3 Explicitly removed from this revision

The chip does not include a four-channel MUX, excitation DAC, waveform generator, synchronous analog demodulator, low-pass integrator, programmable gain, automatic channel scanning, on-chip averaging, threshold detector, bandgap, PLL, radio, or wireless protocol. These features are not required to demonstrate the revised signal path and would create disproportionate integration risk.

2.4 Requirements and open parameters

Table 2: Requirements and closure method.

Item	Status	Direction	Closure method
Channel count	Fixed	One	Direct electrode-return input; no MUX.
ADC resolution	Fixed	8 bits	Static/dynamic simulation and code-density testing.
Excitation	External	Bipolar, low frequency	MCU/board measurement script; amplitude and frequency remain application parameters.
TIA gain	Open circuit value	One fixed R_F	Select from measured sensor-current range and ADC input span.
Sensor load	Proposal baseline	Two-electrode or inter-digitated sensor	Validate first with precision R , C , and RC networks.
Active current	Proposal stretch target	Below 100 μA where feasible	Post-layout block-current budget under a defined conversion rate.
Sleep current	Proposal stretch target	Below 1 μA where feasible	Leakage simulation including pads and always-on digital.
Interface	Fixed	SPI slave plus external clock	MCU timing simulation and board-level transaction test.
Supply/process	PDK dependent	Core and I/O domains per selected process	Use only qualified devices, pads, and ESD structures.

3 System Architecture

3.1 Top-level organization

Figure 1 separates the measurement path from the control path. The sensor drive does not pass through the ASIC. Only the sensor return current enters the TIA. External references set the analog common mode and ADC conversion range.

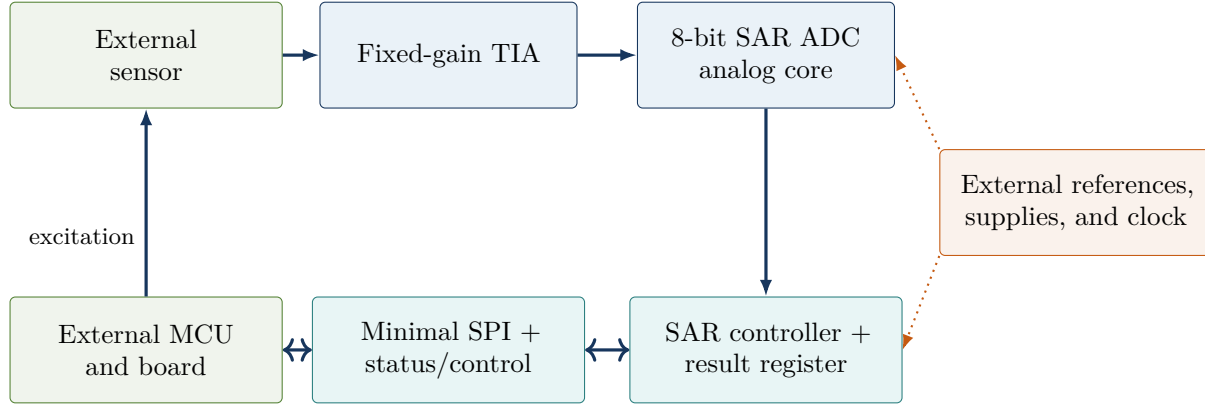


Figure 1: Revised one-channel architecture. The ASIC contains the TIA, ADC, and small digital backend; excitation and signal processing remain external.

3.2 Four paths through the system

1. **Excitation path:** MCU/board $\rightarrow$ driven sensor electrode.
2. **Measurement path:** sensor return current $\rightarrow$ TIA $\rightarrow$ 8-bit SAR ADC $\rightarrow$ result register.
3. **Control path:** MCU $\leftrightarrow$ SPI $\rightarrow$ start/status/result; external clock $\rightarrow$ SAR controller.
4. **Bring-up path:** MCU test routine $\rightarrow$ paired samples $\rightarrow$ subtraction, averaging, and known-load calibration.

This organization is intentionally asymmetric. The ASIC receives the difficult low-current analog signal but leaves flexible waveform generation and numerical processing to equipment that is easy to modify after fabrication.

3.3 Minimal external support

The board-side code is deliberately small: set excitation polarity, wait for settling, request one conversion, poll or observe completion, and read the result through SPI. A short test script may pair opposite-polarity samples and compare known loads. No additional application layer is required for ASIC completion. This boundary keeps the project focused on transistor-level design, RTL, mixed-signal verification, layout, and first-silicon observability.

4 Measurement Method

4.1 Electrical model

For a sensor impedance $Z_S(j\omega)$ and applied excitation $V_{\text{EXC}}(j\omega)$,

$$I_S(j\omega) = \frac{V_{\text{EXC}}(j\omega)}{Z_S(j\omega)}. \quad (1)$$

The TIA converts this current to a voltage around V_{CM} . The ADC returns samples of that voltage; the ASIC does not directly output impedance, moisture, or conductivity.

4.2 External synchronous subtraction

The MCU applies two opposite excitation levels and samples after each half-cycle has settled. Let D_+ and D_- be the ADC codes obtained during positive and negative excitation. The MCU or a minimal test script forms

$$D_{\text{SYNC}} = \frac{D_+ - D_-}{2}. \quad (2)$$

This subtraction rejects much of the static TIA/ADC offset and retains the response that reverses with excitation. Repeating the pair M times gives

$$\overline{D}_{\text{SYNC}} = \frac{1}{M} \sum_{k=1}^M \frac{D_{+,k} - D_{-,k}}{2}. \quad (3)$$

The resulting code is phase-sensitive and impedance-related. Calibration with known loads is still required before converting it to an impedance or agricultural quantity.

4.3 Measurement sequence

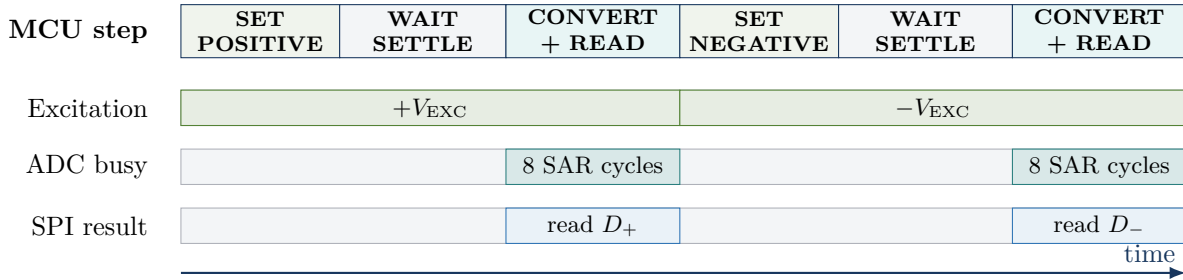


Figure 2: One MCU-controlled sample pair. The ASIC converts individual samples; a minimal external test routine computes D_{SYNC} .

The wait interval must be long enough for electrode, TIA, and board transients to settle. This interval is controlled in firmware, allowing it to change after real sensor measurements without modifying the ASIC.

5 Transimpedance Amplifier

5.1 Function and transfer relation

The TIA maintains the current-sense input near V_{CM} and converts sensor current to voltage. For an ideal resistive feedback path,

$$V_{\text{TIA}}(t) = V_{\text{CM}} - I_{\text{S}}(t)R_{\text{F}}. \quad (4)$$

A feedback capacitor stabilizes the loop and limits high-frequency noise:

$$Z_{\text{F}}(s) = \frac{R_{\text{F}}}{1 + sR_{\text{F}}C_{\text{F}}}. \quad (5)$$

The fixed gain should be selected only after bounding the expected sensor current. If the maximum peak current is $I_{\text{PK,max}}$ and the available linear swing is V_{swing} ,

$$I_{\text{PK,max}}R_{\text{F}} \leq 0.8V_{\text{swing}} \quad (6)$$

is a reasonable first margin rule. The remaining headroom accommodates offset, variation, and switching transients.

5.2 Design priorities

The TIA is the principal analog risk. Its design must address:

- stability across the expected sensor and pad capacitance;
- input-referred noise over the MCU's effective measurement bandwidth;
- output range compatible with V_{REFL} and V_{REFH} ;
- settling after an excitation transition;
- offset and bias current;
- recovery from sensor overload or disconnection;
- enable/disable behavior and sleep leakage;
- pad and ESD leakage relative to the minimum sensed current.

5.3 Fixed gain versus test flexibility

Normal operation uses one feedback network. If schedule and area permit, a test-only alternate feedback resistor may be included to reduce first-silicon risk. It should not expand into a broad programmable-gain feature or require firmware-visible autoranging.

6 8-Bit SAR ADC

6.1 Analog core

The ADC uses a sampling CDAC, comparator, external references, and switch network. During sample, the CDAC acquires the TIA output. During conversion, the SAR controller tests one bit at a time from most significant to least significant. Eight decisions produce one result.

For input range $[V_{\text{REFL}}, V_{\text{REFH}}]$, the ideal code is

$$D = \text{clip} \left(\left\lfloor 256 \frac{V_{\text{TIA}} - V_{\text{REFL}}}{V_{\text{REFH}} - V_{\text{REFL}}} \right\rfloor, 0, 255 \right). \quad (7)$$

The ideal LSB size is

$$V_{\text{LSB}} = \frac{V_{\text{REFH}} - V_{\text{REFL}}}{256}. \quad (8)$$

Eight bits reduce matching, comparator-noise, and digital-control requirements compared with the earlier 10–12 bit concept. The ADC should still be designed so its error is not the dominant limit after TIA noise and sensor variability are included.

6.2 Analog/digital interface

The analog core receives `sample`, CDAC switch controls, and enable. It returns `comparator_out`. The controller must allow adequate CDAC settling and comparator regeneration time for every bit trial. The external reference pins require local decoupling and a board source capable of handling CDAC switching transients.

6.3 ADC verification

Block verification includes offset, gain error, DNL, INL, missing codes, comparator noise, CDAC matching, reference loading, input kickback, conversion time, and energy per conversion. The ADC must also be testable independently of the TIA through a normally disconnected analog test path.

7 Minimal Digital Backend

7.1 Included logic

The backend contains only:

- an 8-state/bit SAR controller plus sample and completion states;
- an 8-bit SAR trial register;
- an 8-bit result register;
- small control/status registers;
- SPI receive/transmit shift logic;
- reset, busy, data-ready, and enable control.

No SRAM, DSP, channel scheduler, frequency divider, accumulator, or autonomous measurement FSM is needed.

7.2 Conversion state sequence

1. **IDLE:** ADC disabled or waiting; previous result retained.
2. **SAMPLE:** enable the ADC and acquire the TIA output.
3. **BIT7 through BIT0:** set the trial bit, wait for settling, sample the comparator, and keep or clear the bit.
4. **STORE:** copy the SAR register to the result register.
5. **DONE:** clear busy and assert data ready.

Any new start request while busy should be ignored and set an error/status bit. Reset must abort a conversion, disable ADC switching, clear status, and return to IDLE.

7.3 Proposed register map

Table 3: Minimal 8-bit register map.

Address	Name	Access	Purpose
0x00	ID	R	Fixed design/revision identifier.
0x01	CONTROL	R/W	Start conversion, ADC enable, soft reset, and test-mode control.
0x02	STATUS	R/W1C	Busy, data ready, overrange, and command-while-busy flags.
0x03	RESULT	R	Latest 8-bit ADC result.
0x04	DEBUG	R	SAR state and selected internal status for bring-up.

SPI mode, edge convention, maximum SCLK, and reset values must be frozen with the MCU test setup. MISO is high impedance whenever chip select is inactive.

8 External References, Power, and Pins

8.1 External analog references

The ASIC receives V_{REFH} , V_{REFL} , and V_{CM} from the board. This removes bandgap startup, temperature curvature, reference buffering, and calibration risk from first silicon. It also makes the TIA and ADC easier to characterize with laboratory supplies.

The chip still needs small local bias circuits for the TIA and comparator. These should be derived from the external reference or a simple externally programmed bias current, depending on the PDK and available pins. The reference choice must be frozen before transistor-level design.

8.2 Power states

Table 4: Simplified power behavior.

State	SPI/registers	TIA	ADC	Purpose
Sleep	Retained	Bias off	Off	Minimize leakage while preserving the latest result.
Analog settle	On	On	Off	Allow the TIA to settle before conversion.
Convert	On	On	On	Sample and perform eight SAR decisions.
Readout	On	As commanded	Off	Return result over SPI.

Bias shutdown is preferred to complex header/footer power gating for this revision. The MCU controls the measurement duty cycle.

8.3 Conceptual pin groups

- **Analog:** sensor-current input, optional TIA/ADC test input, V_{REFH} , V_{REFL} , V_{CM} , analog supply, and analog ground.
- **Digital:** SPI clock, chip select, MOSI, MISO, external SAR clock, reset, enable, optional data-ready, digital supply, and digital ground.
- **Board-only sensor drive:** the excitation connection goes from the MCU/driver to the sensor and is not generated by the ASIC.

9 Calibration and First-Silicon Test

9.1 Calibration flow

The board is first connected to known resistors and RC networks. For a chosen excitation and sampling phase, a minimal test script records D_+ and D_- and computes D_{SYNC} . A simple offset/scale model can begin with

$$R_{CAL} = G(D_{SYNC} - D_{OFF}). \quad (9)$$

More sophisticated mapping should be developed only after real soil or plant electrode data are available.

9.2 Required observability

First-silicon bring-up should support:

- direct observation of the TIA output through a test buffer or switched pad;
- standalone ADC testing with a known external voltage;
- continuous TIA-enable mode for noise and settling measurements;
- visibility of SAR busy/data-ready status;
- result-register reads without starting a new conversion;
- safe reset and analog-disable behavior.

Test switches must be normally open and arranged so their leakage and capacitance do not corrupt normal sensing.

9.3 Bring-up order

1. verify supplies, reset, SPI identity, and quiescent current;
2. verify external references and internal bias points;
3. test the standalone ADC transfer curve;
4. test TIA gain, bandwidth, noise, and settling with precision resistors/current injection;
5. connect the TIA to the ADC and verify end-to-end codes;
6. run positive/negative sample pairs on known RC loads;
7. proceed to controlled soil/plant electrode experiments.

10 Verification and Physical Integration

10.1 Block-level verification

Table 5: Required block evidence.

Block	Minimum evidence before top-level integration
TIA	DC operating point, transimpedance, stability across sensor capacitance, integrated noise, offset, settling, output swing, overload recovery, power, and leakage.
SAR analog core	Sampling accuracy, CDAC settling/matching, comparator noise/offset, DNL, INL, missing-code check, reference loading, conversion energy, and PVT/mismatch results.
Digital backend	SPI protocol tests, reset, start/busy/done behavior, SAR binary-search correctness, illegal command handling, synthesis, timing, and digital power estimate.
Top level	Known-load end-to-end codes, reference/pad interaction, mixed-signal regression, extracted critical paths, safe power sequencing, DRC, LVS, and required foundry checks.

10.2 Mixed-signal regression

The top-level testbench should sweep sensor resistance/capacitance, TIA current, ADC input code, external reference variation, supply, temperature, process, and conversion timing. Assertions must ensure that the SAR result is written only after eight valid comparator decisions and that reset cannot leave the CDAC or analog biases in an unsafe state.

10.3 Layout guidance

Place the sensor input pad close to the TIA summing node. Keep SPI, clocks, and CDAC switching away from that node. Locate the ADC near the TIA output while separating comparator/CDAC references from noisy digital return paths. Use guard rings, dense substrate contacts, shielded sensitive nets, symmetrical CDAC layout, dummy elements, and local reference decoupling according to the PDK.

Each owner lays out and verifies their own block. Top-level floorplanning begins as soon as block interfaces and approximate dimensions are stable.

11 Three-Person Team Structure

The project is divided into three similarly sized signoff blocks. Each person owns design, verification, layout or implementation, documentation, and integration evidence for one block. Ownership means producing a block that another teammate can connect without guessing its ranges, timing, supplies, or failure behavior.

Table 6: Exact primary ownership and required output.

Owner	Owned block	Exact work	Definition of done
Krishna	TIA and sensor interface	Bound sensor current and capacitance; choose the amplifier and bias point; size R_F and C_F ; verify gain, stability, noise, offset, swing, settling, overload, pad parasitics, power, and leakage; create the analog input test path; complete layout and extraction.	Signed-off extracted TIA, plots across PVT and mismatch, sensor/pad assumptions, output-range and settling specification, current budget, and connection model for the ADC.
Vidhu	8-bit SAR analog core	Choose the sampling and CDAC architecture; size capacitors and switches; design the comparator; define common-mode and references; verify acquisition, settling, kickback, comparator noise/offset, transfer curve, DNL/INL, missing codes, energy, PVT, and mismatch; complete matched layout and extraction.	Signed-off extracted ADC analog core, characterization report, reference-loading limits, sampling and decision timing limits, test-input mode, and digital control truth table.
Taara	SAR control, SPI, and top-level digital	Define the SAR sequence and timing; implement SPI, register map, start/busy/done, result storage, reset, enable, and error behavior; build self-checking RTL tests and assertions; run synthesis, timing, and power checks; integrate digital controls and top-level pin connectivity.	Regression-clean RTL and gate-level netlist, protocol and register documentation, timing constraints, synthesis/STA reports, conversion traces, reset-safety evidence, and MCU transaction script.

11.1 Cross-review and integration

Krishna reviews the ADC input range and analog loading. Vidhu reviews the TIA-to-ADC interface and the comparator-to-SAR timing. Taara reviews every control, status, reset, and test connection at the top level. Vidhu reviews Taara's SAR decision sequence, while Taara supplies reproducible digital stimuli for both analog blocks. Final floorplanning, pad-ring decisions, mixed-signal regression, DRC/LVS closure, and the first-silicon bring-up plan are reviewed by all three.

An interface change is accepted only when its schematic or RTL, behavioral model, timing/range specification, and regression test are updated together. This prevents a late change in one block from silently invalidating another person's work.

12 Eight-Week Execution Plan

Table 7: Named eight-week implementation schedule.

Wk.	Gate	Krishna: TIA	Vidhu: SAR	Taara: digital
1	Interfaces frozen	Bound current, capacitance, output span, noise, and settling targets; draft TIA model	Freeze ADC input/reference range, sampling load, control pins, and timing contract	Freeze pins, clocks, SPI mode, register map, SAR handshake, reset states, and top-level test plan
2	Nominal schematics	Select topology and bias; close DC/AC loop; size R_F and C_F	Select CDAC, sampling network, comparator, and reference-switch architecture; nominal simulations	Implement SPI/register RTL and SAR state machine; create unit tests and comparator model
3	Nominal function	Run transient, settling, noise, swing, and overload tests; publish ADC-drive limits	Run transfer, acquisition, settling, comparator, and decision-timing tests; publish control truth table	Reach RTL feature complete; run self-checking conversion tests, protocol corner cases, and synthesis sanity check
4	Midpoint integration	Complete PVT and sensor-capacitance stability sweeps; begin stable layout	Complete PVT, DNL/INL, noise, kickback, and reference-loading sweeps; begin matched layout	Integrate TIA/ADC behavioral models; run end-to-end SPI-code regression; freeze features and interfaces
5	Block signoff	Complete layout, DRC/LVS, extracted feedback loop, mismatch, and current budget	Complete CDAC/comparator layout, DRC/LVS, extraction, mismatch, missing-code, and energy checks	Freeze RTL; complete assertions, synthesis, STA, power estimate, pad controls, and top-level connectivity
6	Top-level assembly	Run extracted block simulations in top-level loading; finalize sensor pad and TIA test routing	Run extracted ADC with digital timing and external-reference models; finalize ADC test path	Assemble mixed-signal top level; run gate-level checks, MCU transaction script, power/reset tests, and floor-plan connectivity
7	Closure	Resolve extracted TIA/top-level coupling, stability, leakage, and physical issues	Resolve extracted ADC/reference/coupling and timing issues; complete final characterization plots	Run full regression and top-level timing; coordinate DRC/LVS/ERC/antenna and fix digital/top-connectivity issues
8	Release candidate	Archive signed-off TIA source, extracted results, plots, limits, and bring-up procedure	Archive signed-off ADC source, extracted results, characterization, timing, and bring-up procedure	Archive RTL, netlist, reports, and tests; close top level; assemble checksums, pinout, register guide, and team release package

12.1 Weekly decision gates

- **End of Week 1:** no unresolved ownership, pin, voltage-domain, range, clock, or handshake questions.
- **End of Week 3:** every block meets nominal function and has a written interface contract.
- **End of Week 4:** no new features; fixed gain and 8-bit resolution remain frozen.
- **End of Week 6:** extracted analog blocks, synthesized digital, pad controls, and top-level connectivity are integrated.
- **End of Week 8:** physical checks, end-to-end known-load demonstration, and archived hardware evidence are complete.

13 Risk Register

Table 8: Primary remaining risks.

Risk	Prob.	Impact	Mitigation	Fallback
Unknown sensor current	High	High	Measure representative RC loads early; leave TIA test access; maintain headroom	Change board excitation amplitude or feedback option.
TIA instability	Medium	High	Include worst-case pad/sensor capacitance; tune C_F ; verify phase margin	Lower excitation frequency or increase compensation.
ADC/TIA range mismatch	Medium	High	Freeze common-mode/reference plan and behavioral range model in Week 1	Adjust external references or excitation amplitude.
CDAC/reference kickback	Medium	Medium	Local decoupling, sampling isolation, extracted transient checks	Slow SAR clock; strengthen external reference drive.
Digital coupling into TIA	Medium	High	Floorplan separation, quiet sampling phase, shielding and return-path planning	Stop SPI during sampling; slow clocks.
Top-level interface drift	Medium	High	Freeze voltage ranges, timing, controls, and behavioral models in Week 1; regression-check interfaces	Reject late interface changes unless all three owners approve and tests remain green.
Sleep-current miss	Medium	Medium	Leakage budget including pads; explicit bias shutdown	Relax sleep target while reporting measured duty-cycle energy.
Physical closure late	Medium	High	Begin layout in Week 4; run DRC/LVS continuously	Remove optional test routing before core path.

14 What Must Not Be Added Back

The simplified design succeeds only if its boundary remains stable. During the two-month window, the team should not reintroduce multiple channels, an excitation DAC, analog demodulation, programmable gain, an on-chip bandgap, additional ADC resolution, automatic averaging, wireless, or a PLL. Any of these changes would create new analog interfaces and verification scope.

If schedule pressure still develops, simplify in this order:

1. remove optional debug registers;
2. use a dedicated start pin instead of a writable control register;
3. reduce optional analog test routing while retaining one TIA observation point;
4. retain the TIA, 8-bit SAR conversion, one result register, reset safety, and readable output.

15 Conclusion

The revised ASIC is intentionally compact but still meaningfully mixed signal. Its TIA addresses the difficult low-current sensor interface; its 8-bit SAR ADC performs on-chip analog-to-digital conversion; and its digital backend executes the SAR search, stores a result, and communicates through SPI. The external MCU supplies excitation and flexibility, including phase-sensitive subtraction, averaging, calibration, and application interpretation.

This scope produces three clear and similarly sized ownership areas: Krishna signs off the TIA and sensor interface, Vidhu signs off the SAR analog core, and Taara signs off the SAR controller, SPI, and top-level digital integration. The main success criterion is not feature count. It is a fabricated, observable signal path whose behavior can be explained from electrode current to SPI code and demonstrated with known loads.

A Conceptual Internal Signal Contract

Table 9: Internal and top-level control signals.

Signal	Domain	Source → destination	Meaning
<code>start</code>	Digital	SPI control → SAR FSM	Begins one sample/conversion when idle.
<code>sample</code>	Digital	SAR FSM → ADC core	Connects the CDAC to the TIA output for acquisition.
<code>dac[7:0]</code>	Mixed	SAR register → CDAC	Trial code for the current SAR decision.
<code>comp_out</code>	Mixed	Comparator → SAR FSM	Comparator result sampled after settling.
<code>busy</code>	Digital	SAR FSM → status	High from sample start through result storage.
<code>data_ready</code>	Digital	SAR FSM → status/pin	Latest result is valid and stable.
<code>result[7:0]</code>	Digital	SAR FSM → result register	Completed conversion code.
<code>tia_enable</code>	Digital	Control → TIA bias	Wakes the front end; requires a defined settling delay.
<code>adc_enable</code>	Digital	SAR FSM → ADC bias	Enables comparator/CDAC activity only for conversion.

B Pre-Tapeout Checklist

- ☐ Sensor-current and capacitance range are documented with a conservative margin.
- ☐ External reference, common-mode, supply, clock, pad, and ESD assumptions are frozen.
- ☐ TIA passes stability, noise, settling, overload, power, leakage, PVT, mismatch, and extracted checks.
- ☐ ADC passes DNL/INL, missing-code, noise, timing, reference-loading, PVT, mismatch, and extracted checks.
- ☐ Digital backend passes SPI, reset, start/busy/done, binary-search, illegal-command, synthesis, and timing tests.
- ☐ End-to-end regression maps known TIA input currents to expected SPI result codes.
- ☐ A minimal MCU transaction script starts conversions, reads status/results, and handles reset or error cases.
- ☐ Positive/negative sample pairs from known resistive and RC loads produce repeatable calibrated results.
- ☐ Analog test paths are observable and isolated during normal operation.
- ☐ Power-up, power-down, abort, and reset states are safe.
- ☐ Pad ring, supply returns, floorplan, DRC, LVS, ERC/antenna, and extracted critical paths are closed.
- ☐ Source, models, scripts, reports, plots, checksums, and bring-up documentation are archived.

References

- [1] Senior Design Team, *Mixed-Signal Impedance Sensing ASIC for Battery-Free Agricultural IoT Nodes*, Senior Design Project Proposal, 2026.